

# Comparative Analysis of AI-Assisted and Manual Data Labeling Accuracy Across Scientific Datasets

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**Abstract**—The unprecedented proliferation of high-dimensional data across scientific disciplines has established data annotation as the primary bottleneck in the development, validation, and deployment of supervised machine learning systems. For decades, the foundational bedrock of artificial intelligence has been the “ground truth” provided by manual human annotation. In highly specialized domains such as computational pathology, single-cell transcriptomics, clinical trial management, geospatial remote sensing, and astrophysics, this manual process demands significant domain expertise. Consequently, it is prohibitively expensive, inherently unscalable, and notoriously susceptible to human fatigue, cognitive bias, and inter-observer variability. The historical assumption that expert manual labeling represents an infallible gold standard has been increasingly challenged by empirical evidence demonstrating that even highly curated benchmark datasets contain substantial rates of label noise and systematic bias. To address these critical inefficiencies, the scientific community is rapidly shifting toward AI-assisted data labeling. This paradigm operates on a Human-in-the-Loop (HITL) framework, wherein machine learning algorithms—ranging from foundation models and large language models (LLMs) to active learning heuristics and zero-shot classifiers—provide initial annotations, segmentations, or localized bounding boxes.

**Index Terms**—Data Labeling, AI-Assisted Annotation, Human-in-the-Loop, Ground Truth, Scientific Datasets

## I. INTRODUCTION

These algorithmic suggestions are subsequently reviewed, refined, or adjudicated by human annotators. This comprehensive comparative analysis examines the accuracy, efficiency, statistical reliability, and cognitive implications of AI-assisted versus manual data labeling across diverse scientific datasets. By synthesizing empirical evidence from medical imaging, genomics, clinical text processing, and astronomy, this report elucidates the transformative potential of hybrid intelligence while critically evaluating the emergent risks of automation bias, out-of-distribution shifts, and the propagation of label noise [1].

As illustrated in Fig. 1, the integration of human expertise remains paramount despite algorithmic efficiency.

## II. THEORETICAL AND STATISTICAL FRAMEWORKS FOR GROUND TRUTH

Before directly comparing manual and AI-assisted modalities, it is necessary to establish the mathematical and statistical frameworks used to define “accuracy” in contexts where

Conceptual Architecture: Human-in-the-Loop AI-Assisted Scientific Data Labeling Pipeline

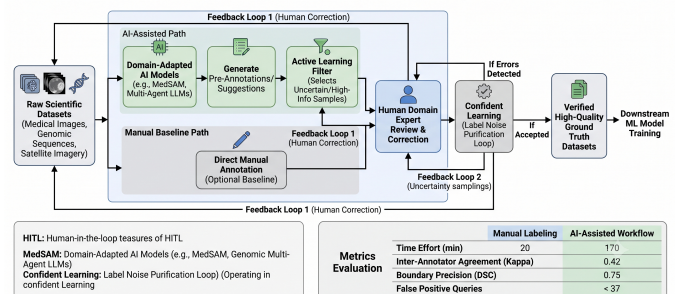


Fig. 1. Conceptual Architecture of the Human-in-the-Loop AI-Assisted Scientific Data Labeling Pipeline.

absolute objective truth is fundamentally inaccessible. In scientific datasets, ground truth is typically established through consensus among multiple domain experts. The reliability of this consensus is quantified using Inter-Annotator Agreement (IAA) metrics, which evaluate the consistency with which multiple annotators assign categories, segment boundaries, or parse textual relationships in a shared dataset.

### A. Rethinking Annotation as a Measurement Process

Machine learning research has conventionally treated all annotator disagreement as noise. However, the Human Label Variation (HLV) framework posits that annotation is a highly complex measurement process where label variation stems from four distinct sources: instance difficulty, annotator bias, situational noise, and relational alignment. Decomposing these outcomes reveals that what algorithms perceive as “noise” is often valid, subjective medical or scientific disagreement over ambiguous phenomena. When researchers blindly enforce consensus (e.g., via simple majority voting) on ambiguous items, they obscure these critical distinctions and limit the downstream model’s ability to navigate edge cases [1]. A study involving eleven critical care consultants annotating Intensive Care Unit (ICU) data revealed only “fair” agreement (Fleiss’  $\kappa = 0.383$ ) internally, and “minimal” agreement ( $\kappa = 0.255$ ) when externally validated against out-of-network datasets, demonstrating that a singular “super expert” rarely exists in acute clinical settings [2].

### B. Quantifying Inter-Annotator Agreement

To rigorously evaluate annotation quality, researchers rely on metrics that correct for chance agreement, distinguishing between true scientific consensus and random overlap [3]. Cohen’s Kappa ( $\kappa$ ) is the standard metric for binary or categorical classification tasks involving two annotators. It is formulated mathematically as:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \quad (1)$$

Where  $P_o$  represents the relative observed agreement among raters, and  $P_e$  represents the hypothetical probability of chance agreement in (1). In complex medical annotation tasks, such as classifying oral potentially malignant disorders from intraoral images, manual expert agreement often reaches  $\kappa = 0.90$  indicating excellent reliability.

For datasets requiring annotation by more than two experts, Fleiss’ Kappa extends the mathematical foundation of Cohen’s Kappa to assess the consistency of multiple raters evaluating the same items simultaneously. Krippendorff’s Alpha ( $\alpha$ ) provides an even more robust and flexible formulation, accommodating nominal, ordinal, interval, and ratio data, while successfully handling missing values—a common occurrence in highly sparse scientific datasets.

In computer vision tasks, particularly medical image segmentation and satellite remote sensing, accuracy cannot be evaluated using categorical agreement alone; it requires spatial overlap metrics. The Dice Similarity Coefficient (DSC) and Intersection over Union (IoU) serve as the primary standards:

$$\text{DSC} = \frac{2|X \cap Y|}{|X| + |Y|} \quad (2)$$

$$\text{IoU} = \frac{|X \cap Y|}{|X \cup Y|} \quad (3)$$

Where  $X$  and  $Y$  represent the spatial masks generated by different human annotators or by an AI model against a human reference, demonstrated in (2) and (3). To aggregate varying human annotations into a single, unified ground truth mask for AI training, algorithms such as Simultaneous Truth and Performance Level Estimation (STAPLE) are deployed. STAPLE computes a probabilistic estimate of the true segmentation based on the historical performance level and bias of each individual annotator, yielding a highly resilient training target.

### III. MEDICAL IMAGING AND COMPUTATIONAL PATHOLOGY

The integration of artificial intelligence into medical image annotation has demonstrated profound improvements in both processing speed and boundary precision. Medical imaging requires pixel-perfect accuracy, as downstream clinical models utilize these labels for oncology diagnostics, surgical planning, and anomaly detection.

TABLE I  
INTER-ANNOTATOR AGREEMENT (IAA) METRICS

| IAA Metric           | Mathematical Characteristics | Primary Application                      | Strengths in Scientific Labeling                                          |
|----------------------|------------------------------|------------------------------------------|---------------------------------------------------------------------------|
| Cohen’s Kappa        | Binary/Categorical           | Corrects for chance agreement            | Standard for dual-annotator verification (e.g., textual triage).          |
| Fleiss’ Kappa        | Multi-rater Categorical      | Generalization of Cohen’s for $n$ raters | Ideal for crowdsourced or multi-expert panel consensus.                   |
| Krippendorff’s Alpha | Complex/Mixed Data           | Handles missing data and various scales  | Highly robust for sparse datasets with incomplete overlapping reviews.    |
| STAPLE               | Spatial Segmentation         | Expectation-Maximization algorithm       | Aggregates multiple bounding boxes/masks into a true probabilistic label. |
| Dice / IoU           | Spatial Overlap              | Harmonic mean of precision and recall    | Standard for evaluating AI boundaries against human baseline masks.       |

#### A. Foundation Models in Radiology: SAM and MedSAM

The Segment Anything Model (SAM), a foundation model trained on over one billion mask annotations in the natural image domain, catalyzed a shift from specialist segmentation networks to prompt-driven generalist models [4]. However, the direct application of zero-shot SAM to medical imaging often yields suboptimal accuracy due to the vast domain gap between natural photography and complex radiological scans [4].

To bridge this gap, domain-adapted models such as MedSAM have been fine-tuned on extensive medical datasets encompassing 1,570,263 image-mask pairs across 10 imaging modalities and over 30 cancer types [5]. Comparative studies evaluating zero-shot inference for lumbar spine MRI segmentation reveal a distinct hierarchy in annotation accuracy. When segmenting intervertebral discs (IVDs) and vertebral bodies (VBs), standard SAM achieved DSCs of 0.64 and 0.83, respectively [6]. MedSAM significantly outperformed the baseline with DSCs of 0.79 and 0.88 [6]. Despite this marked improvement, fully supervised, task-specific architectures like nnU-Net continue to represent the upper boundary of clinical accuracy, achieving a DSC of 0.99 for the same anatomical structures [6]. Similarly, evaluations on the BraTS 2023 adult glioma and pediatrics dataset indicated that while SAM and SAM 2 can achieve high Dice scores (up to 0.894) when

provided with perfectly precise bounding box prompts, nnU-Net remains dominant due to the practical impossibility of humans consistently providing such perfect prompts during routine clinical workflows [7].

Furthermore, utilizing Granular Box Prompts (GB-SAM)—where clinicians draw localized bounding boxes around regions of interest rather than outlining complex polygons—drastically minimizes human effort while raising accuracy. In histopathology datasets (e.g., CRAG), GB-SAM achieved a DSC of 0.885 utilizing only 25% of the training data, effectively outperforming traditional U-Net architectures (DSC 0.857) trained on identical subsets [8].

### B. Computational Pathology and Annotation Throughput

In computational pathology, Whole Slide Images (WSI) contain billions of pixels, making manual annotation excessively labor-intensive and prone to fatigue. A benchmark study analyzing different annotation peripherals and methodologies demonstrated the extreme efficiency gains of AI-assisted workflows [7]. Pure manual annotation using a standard computer mouse averaged 29.9 minutes per slide with a high 24% inter-observer variability [1]. Touchpad annotation was the least efficient, averaging 47.5 minutes with a 45% variability [9]. Conversely, a semi-automated, AI-assisted approach combining initial manual interactions with SAM reduced the average annotation time to 13.6 minutes while simultaneously achieving the lowest inter-observer variability at just 2% [9].

Similarly, AI-collaborative labeling (AICL) frameworks deployed for complex volumetric tasks, such as CT intracavitary hemorrhage segmentation, demonstrate substantial geometric improvements. AICL reduced the manual time effort by 2.8-fold compared to standard clinical staff editing and 8.7-fold compared to expert radiologist labeling from scratch [10]. The qualitative Likert grades awarded to the final annotations were significantly higher for AI-assisted labels (8.4) compared to purely manual expert labels (7.8), suggesting that algorithmic assistance uncovers boundary details that human fatigue naturally obscures over long sessions [10].

### C. Hybrid Intelligence and Uncertainty-Driven Workflows

To optimize the balance between automated efficiency and human clinical judgment, "HybridMS" (Hybrid Medical Segmentation) frameworks have been introduced. These systems rely on uncertainty-driven feedback mechanisms that evaluate the AI's confidence in its own segmentation. If the model determines a region is statistically ambiguous (e.g., poorly contrasted tumor boundaries), it selectively triggers a human clinician for input [11]. In a lung segmentation trial for tuberculosis detection on chest X-rays, HybridMS achieved a Dice score of 0.9538 (compared to a baseline MedSAM of 0.9435), while reducing standard annotation time by 82% and challenging case annotation time by 60% [11]. This selective intervention maximizes throughput without compromising regulatory or clinical safety standards.

TABLE II  
PERFORMANCE AND EFFICIENCY ACROSS MODALITIES

| Modality / Architecture | Clinical Application       | Performance Metric (Dice) | Efficiency Gains over Manual Baseline    |
|-------------------------|----------------------------|---------------------------|------------------------------------------|
| Zero-Shot SAM           | Lumbar MRI                 | DSC: 0.64-0.83            | High (Requires exact prompts)            |
| MedSAM                  | Lumbar MRI                 | DSC: 0.79-0.88            | High (Lower prompt dependency)           |
| nnU-Net (Supervised)    | Lumbar/Glioma              | DSC: 0.99                 | None (Requires massive pre-labeled data) |
| GB-SAM (Box Prompt)     | Histopathology (CRAG)      | DSC: 0.885                | Achieved with only 25% of required data  |
| Semi-Automated SAM      | WSI Pathology              | 2% Variability            | Time reduced from 29.9 min to 13.6 min   |
| HybridMS (Uncertainty)  | Chest X-ray (Tuberculosis) | DSC: 0.9538               | Annotation time reduced by 82%           |

## IV. GENOMICS, SINGLE-CELL TRANSCRIPTOMICS, AND CLINICAL TEXT PROCESSING

Beyond visual segmentation, a vast proportion of scientific datasets consist of unstructured clinical text, concomitant medication records, adverse event narratives, and high-dimensional genomic sequences.

### A. Single-Cell RNA Sequencing (scRNA-seq) Annotation

Single-cell RNA sequencing (scRNA-seq) provides unprecedented resolution into cellular heterogeneity, allowing researchers to build highly detailed transcriptomic atlases. Traditionally, cell type annotation has relied heavily on manual intervention. Domain experts compare differentially expressed genes in computationally generated clusters against canonical marker genes documented in literature [12]. This manual approach is highly subjective, labor-intensive, and lacks standardized criteria across different laboratories, leading to fragmented and incompatible datasets [12].

The introduction of specialized Large Language Models and multi-agent AI systems has fundamentally restructured bioinformatics pipelines. Advanced benchmarking of leading LLMs (e.g., Kimi-k2, GPT-5, Claude-4.1, Grok-4) across 34 diverse human and mouse datasets demonstrated that AI models profoundly outperform traditional computational methods (such as SingleR, Cellmarker2.0, and ScType), particularly in their ability to resolve fine-grained, rare cell subtypes [12]. These capabilities are integrated into platforms like DeepCellSeek, which leverage statistical significance and log-fold change rankings to optimize the prompts fed into LLMs [12].

### B. Clinical Trial Data Cleaning and Extraction

In pharmaceutical drug development, cleaning clinical trial data represents a severe operational bottleneck. The manual reconciliation of these datasets is highly susceptible to human error due to repetitive task fatigue and the inability to scale with exponentially growing data volumes [13].

In a controlled experimental study involving experienced medical reviewers, the implementation of "Octozi"—an AI-assisted platform combining Large Language Models with domain-specific clinical heuristics—yielded unprecedented improvements [13]. The AI-assisted workflow increased data cleaning throughput by 6.03-fold [13]. More importantly, the system caused cleaning errors to plummet from a manual baseline of 54.67% to just 8.48%, representing a 6.44-fold improvement in accuracy [13].

## V. ASTROPHYSICS AND GEOSPATIAL REMOTE SENSING

In disciplines characterized by massive planetary or cosmic scale, such as Earth observation and astrophysics, the sheer volume of high-resolution imagery necessitates automated pipelines. However, labeling this data requires nuanced spatial and morphological reasoning that challenges basic computer vision architectures.

### A. Galaxy Morphology Classification and Green AI

The crowdsourced citizen science project Galaxy Zoo historically demonstrated the power of distributed manual annotation, where volunteers classified millions of galaxies into morphological categories (e.g., spiral, elliptical, merging) through complex decision trees [14]. As telescope capabilities advance—such as with the Rubin Observatory Legacy Survey of Space and Time (LSST) and the Euclid space mission—the volume of data renders human crowdsourcing mathematically insufficient [14].

Deep learning models pre-trained on Galaxy Zoo labels, such as the Zoobot architecture, have successfully automated the detection of complex morphological phenomena. In the identification of lopsided spiral galaxies versus symmetric galaxies, fine-tuned Zoobot models achieved a testing accuracy of 87% against expert-verified ground truth, successfully labeling datasets that would take human crowds years to process [15]. Similarly, YOLOv5 architectures have been deployed to detect edge-on galaxies, achieving detection rates approaching 97% over vast survey data like Pan-STARRS [16]. Modern approaches are transitioning away from astronomy-specific supervised models toward general-purpose Vision-Language Models (VLMs) and Visual Question Answering (VQA) frameworks.

## VI. THE PSYCHOLOGY AND ERGONOMICS OF HUMAN-IN-THE-LOOP ANNOTATION

While the quantitative benefits of AI-assisted labeling are statistically undeniable across scientific domains, the insertion of AI into the human decision-making loop introduces complex cognitive and psychological phenomena.

### A. Annotation Fatigue and Cognitive Load

Manual labeling over long durations induces severe annotation fatigue, leading to a demonstrable decline in data quality. Fatigue manifests as mental exhaustion stemming from high cognitive load, repetitive visual scanning, and subjective stress when dealing with ambiguous data guidelines [17]. Empirical observations indicate that prolonged manual annotation can drive error rates as high as 30%, rendering the data unusable for safety-critical clinical applications or precise scientific tracking [18].

### B. Automation Bias and The Anchoring Effect

The central paradox of AI assistance is that as algorithms become more highly accurate, human overseers become more complacent. This phenomenon, known as automation bias or the anchoring effect, occurs when human annotators overly rely on AI-generated suggestions, subsequently failing to detect or correct algorithmic errors [19]. The human effectively anchors their judgment to the AI's initial prediction, known quantitatively as the "Weight of Advice" (WoA) [20].

TABLE III  
COGNITIVE MECHANISMS AND MITIGATION STRATEGIES

| Cognitive Phenomenon   | Mechanism                                                 | Mitigation Strategy                                                   | Impact on Annotation Quality                                        |
|------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------|
| Annotation Fatigue     | Mental exhaustion from repetitive, high-load manual tasks | AI pre-labeling, task rotation, ergonomic UI design                   | Increased random errors, dropped boundaries, slower throughput      |
| Automation Bias        | Complacent trust in highly accurate AI systems            | Injecting synthetic errors to test vigilance, uncertainty routing     | Systematic encoding of AI hallucination into ground truth           |
| Anchoring Effect (WoA) | Judgment skewed by the initial AI hypothesis              | Blind review of low-confidence predictions, removing time constraints | Inability to correct subtle boundary deviations under time pressure |

## VII. MITIGATING LABEL NOISE: ACTIVE LEARNING, CONFIDENT LEARNING, AND OOD DETECTION

To maximize the efficiency of the human-AI partnership and counteract cognitive biases, advanced machine learning paradigms govern the flow of data to human annotators.

### A. Active Learning and Uncertainty Sampling

Active learning operates on the principle that not all data points contribute equally to model training [21]. Rather than randomly selecting data for human review, the algorithm actively queries the human annotator to label only the most informative or uncertain instances. This targeted methodology

reduces the total annotation budget—often by 99% in dense corpora—while maintaining or exceeding the accuracy of a model trained on a fully, randomly sampled dataset [22].

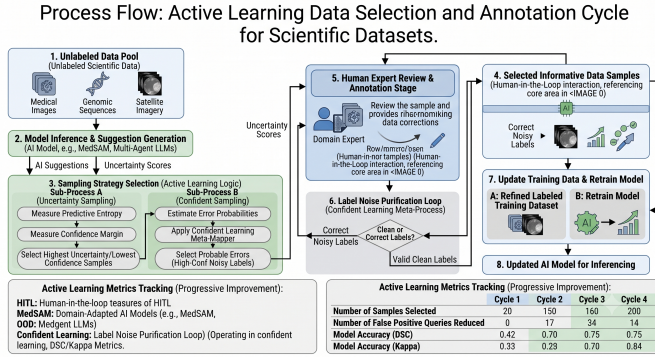


Fig. 2. Process Flow of the Active Learning Data Selection and Annotation Cycle.

As shown in Fig. 2, advanced iterations, such as the ActiveAED (Active Annotation Error Detection) framework, interleave error detection directly into the active learning loop. Utilizing the Area-under-the-Margin (AUM) metric, ActiveAED continually queries humans for error corrections during the prediction loop [23].

### B. Confident Learning and Label Noise Purification

Even with meticulous human-in-the-loop oversight, real-world datasets inevitably contain label noise. Confident Learning (CL) is a model-agnostic meta-learning framework explicitly designed to identify, characterize, and remove label errors from datasets before model training [21].

## VIII. CONCLUSION

The transition from purely manual to AI-assisted data labeling marks a fundamental structural evolution in the production of scientific knowledge. In medical imaging, foundation models like MedSAM, coupled with uncertainty-driven HybridMS architectures, significantly reduce the temporal burden of pixel-wise segmentation while preserving high levels of diagnostic fidelity. In genomics and transcriptomics, large language models organized via multi-agent systems are rapidly replacing subjective manual literature reviews.

However, this comprehensive analysis reveals that AI assistance is not a panacea for data quality. The integration of algorithmic pre-labeling introduces acute psychological variables—specifically, automation bias, anchoring effects, and the risk of uncritical complacency among human reviewers. While AI tools successfully mitigate annotation fatigue and decrease inter-observer variability, their ultimate efficacy is strictly tethered to the cognitive engagement of the human in the loop. The optimal scientific annotation strategy relies heavily on Active Learning and Confident Learning frameworks that purposefully measure uncertainty.

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